

helix-core (Rust client/ connector core)

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Revision: 2 Last modified: 2026-07-04T12:00:00Z

Master technical specification — **Volume 4 (Clients)**, nano-detail deep-dive. This document **deepens** the helix-core (Rust client/connector core) part of the pass-1 client overview [04_UI §0/§1/§5, 04_ARCH §5.1] into an implementation-ready specification of **the Rust workspace that the FFI consumes**: the cargo crate layout (helix-transport, helix-wg, helix-ffi, the orchestrator+reconciler in helix-core), the size/memory build strategy (opt-level=z · LTO · panic=abort · strip), the network-map reconciler (consume RouteMap/MapDelta → bring WG peers up/down, update AllowedIPs), the kill-switch + DNS state machine, the --mode=connector posture sharing the same core, and **how the core sits beneath the FFI**.

Boundary with sibling docs. This document is the *client-core* counterpart of the data-plane internals owned by Volume 2: the **orchestrator three-loop, the TunnelStatus broadcast, the reconciler diff/apply, the backoff machine, and the kill-switch/DNS coupling are specified in depth by** [v02-data-plane/orchestrator-and-state.md](#) (cited [v02-orch §N]) and the carrier seam by [v02-data-plane/transport-trait.md](#) (cited [v02-tx §N]). This document **owns** the *packaging + FFI seam* of that core: the cargo workspace, the per-platform library artifacts, the build/memory profile, the helix-ffi crate surface, and the FFI-driven *entry points* into the orchestrator/reconciler/kill-switch (it drives them; v02 defines their bodies). It does **not** re-specify the three loops, the Tunn crypto, the ladder algorithm, or the per-carrier wire formats. The FFI surface and the TunnelStatus enum here are **kept byte-consistent with** [v02-orch §4.1, §5] and the FFI sketch in [04_UI §5 / doc 03 §3].

Evidence base. Citations inline by id: [04_UI] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_P0 §N] = ...Phase0-Spike.md; [research-flutter_ffi] / [research-ios_android] = the cited research digests; [v02-orch §N] / [v02-tx §N] = the Volume-2 data-plane nano-docs; [SYN §N] = cross-document synthesis. Any claim not grounded in the evidence base is tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

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0. Position, ownership, and invariants

helix-core is the **bounded-memory Rust data-plane core** shared by three consumers — Helix Access (client), Helix Connector (advertise/route), and the gateway edge — from *one* workspace [04_ARCH §5.1/§5.5, SYN §5, v02-tx §0]. On the client side it is consumed **through an FFI**: flutter_rust_bridge (frb) v2 emits the Dart glue, UniFFI emits the Swift/Kotlin glue where a native OS extension links the core directly [04_UI §5, research-flutter_ffi §1/§2]. This document is the nano-detail of *that core as a shippable, FFI-consumable artifact*.

0.1 What this document owns

#	Owned here	Deepened from
W1	The cargo workspace — crates, crate-type, feature flags, dependency DAG (§1)	[04_ARCH §11, SYN §6]

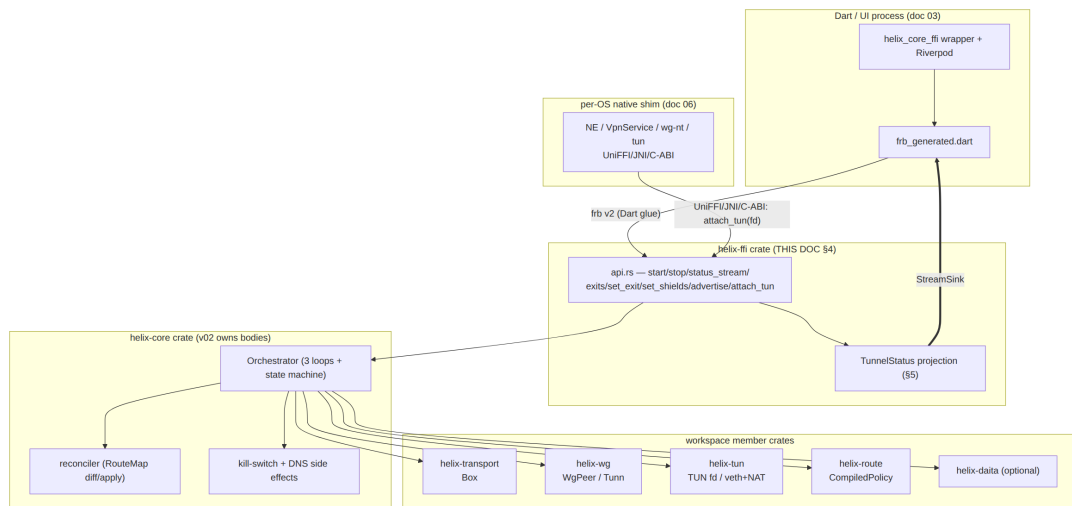
#	Owned here	Deepened from
W2	The per-platform library artifacts (staticlib/cdylib/xcframework/.so/.a/.dll) and how each shim links them (§2)	[research-flutter_ffi §3, research-ios_android §2/§3]
W3	The size/memory build profile (opt-level=z · LTO · codegen-units=1 · panic=abort · strip · iOS build-std) + budgets (§3)	[research-ios_android §1/§2, 04_P0 §6]
W4	The helix-ffi crate surface — api.rs (start/stop/status_stream/exits/set_exit/set_shields/advertise/attach_tun/detach_tun), the mirrored value types (§4)	[04_UI §5, doc 03 §3, v02-orch §2.1]
W5	The TunnelStatus projection — core 5-variant (v02) $\Rightarrow$ Dart-facing FFI enum, the StreamSink pump, receiver contract (§5)	[v02-orch §4.1, doc 03 §3.1]
W6	The FFI-driven entry points into the orchestrator/reconciler/kill-switch — attach_tun, apply_map, set_shields — and the C-ABI TUN-fd seam (§6, §7, §8)	[v02-orch §2.1/§6/§8, research-ios_android §3]
W7	The --mode=connector packaging — one binary, mode flag, headless daemon endpoint (§9)	[v02-orch §9, 04_ARCH §5.4]

0.2 What this document does NOT own

- The orchestrator three loops, the backoff machine, the broadcast sender body — [v02-orch §3/§4/§7]. This doc drives them through helix-ffi.
- The Transport trait, dial(), per-carrier wire formats — [v02-tx §2/§4]. The core holds opaque Box<dyn Transport> handles.
- WireGuard crypto / the Tunn state machine (helix-wg) — doc 01 §4.
- The Dart helix_core_ffi wrapper, Riverpod, TunnelPlatform Dart contract, and every pixel of UI — doc 03 §3.2/§4/§8 [04_UI §4/§5/§6].
- The WatchNetworkMap protobuf that *delivers* the RouteMap — doc 02. The core consumes the map's *shape*; a file-watch on map.json is the Phase-0 stand-in [04_P0 §10].

0.3 Invariants this document inherits and enforces at the FFI seam

#	Invariant	Source	FFI-seam enforcement
F-I1	The core owns truth about protection state; the UI is a pure function of the status stream.	[04_UI §4.2 CI2]	The FFI exposes <i>only</i> <code>status_stream()</code> ; there is no <code>is_connected()</code> poll getter (§5.4).
F-I2	The core never opens the OS tunnel. The shim creates the TUN and hands the core a packet fd / pump via <code>attach_tun</code> (the one place crypto+transport live stays in Rust).	[04_UI §5 CI1, v02-tx I4]	<code>start()</code> does logic only; <code>attach_tun(fd)</code> is the sole packet-path entry (§6).
F-I3	One core, three consumers, no forked bodies — client/connector differ only by Mode.	[v02-orch O-I13]	<code>CoreMode</code> is a <code>start()</code> parameter; §9 adds no second loop body.
F-I4	Fail-closed by construction — kill-switch/DNS side effects are a function of <code>OrchState</code> transitions, surviving crash.	[v02-orch O-I11]	The FFI cannot open the firewall directly; only a state transition does (§8).
F-I5	No-logging / no-secret-leak by construction.	[v02-tx I5, §11.4.10]	<code>SecretBytes</code> redacts+zeroizes; the FFI never returns a secret; the status stream carries only aggregate state (§5.5).
F-I6	iOS-NE memory ceiling is the hardest budget and the reason the core is Rust not Go; every build/runtime decision defers to it.	[04_ARCH §5.6 CI4, research-ios_android §1]	§3 build profile + §3.3 runtime budget.



1. The cargo workspace & crate layout

helix-core/ is a **single cargo workspace** (snake_case, flat submodule per §11.4.28/.29/.74) [v02-tx §0, SYN §6/§9]. Each member is decoupled and independently testable; only helix-ffi and the binary depend on the OS/FFI.

```

helix-core/
vasic-digital repo)
├─ Cargo.toml
[profile.*] (§3)
├─ upstreams/
(§11.4.36)
├─ crates/
|   └─ helix-transport/
+ impls [v02-tx]
|   └─ crate-type = ["rlib"]
|   └─ helix-wg/
(boringtun wrap) [doc 01 §4]
|   └─ crate-type = ["rlib"]
|   └─ helix-tun/
(client fd / connector veth+NAT)
|   └─ crate-type = ["rlib"]
|   └─ helix-route/
verdict map [doc 01 §6-§7]
|   └─ crate-type = ["rlib"]
|   └─ helix-daita/
(maybenot) [doc 01 §9]
|   └─ crate-type = ["rlib"]
|   └─ helix-core/
state machine [v02-orch]
|   └─ crate-type = ["rlib"]
the loops, NOT exported directly)
|   └─ helix-ffi/

```

the Rust workspace (own

[workspace] members + shared

install_upstreams recipes

L2 carriers: Transport trait

WG crypto core: WgPeer

TUN device abstraction

RouteMap, CompiledPolicy,

optional L2.5 shaping

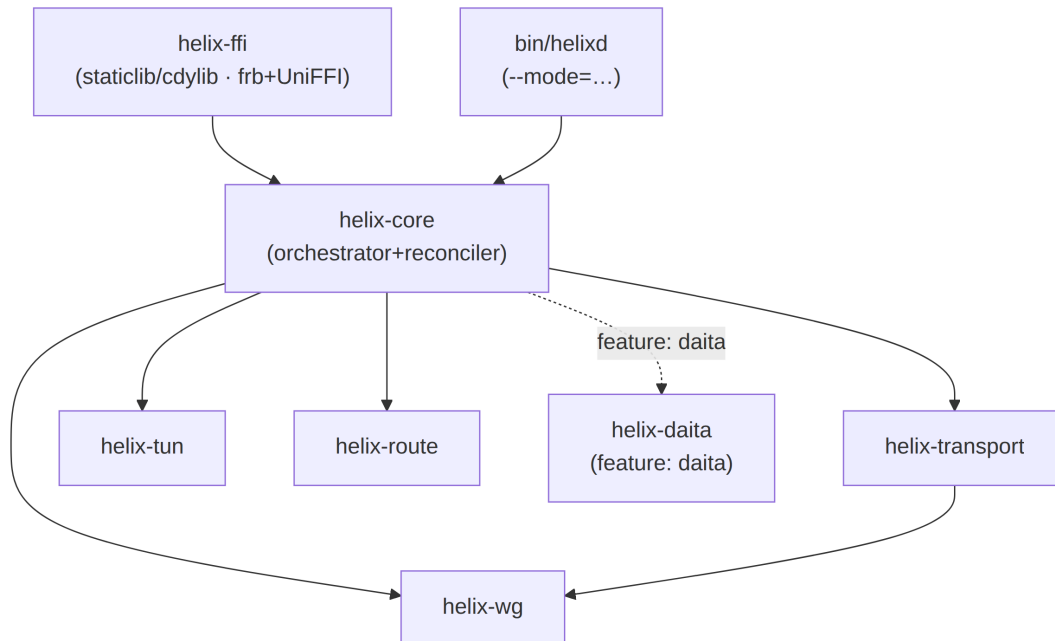
ORCHESTRATOR + RECONCILER +

(the "conductor" – owns

THE FFI SURFACE (THIS DOC §4)

```
|      └─ crate-type = ["staticlib", "cdylib"]    # platform-
selected at build (§2)
└─ bin/
    └─ helixd.rs                                # the daemon: --mode={client|
connector}, no Flutter (§9)
```

1.1 Dependency DAG (acyclic, FFI at the apex)



Rules: (1) **no crate depends on helix-ffi** — the FFI is a sink, never a dependency, so the same `helix-core` links into the edge binary with zero FFI weight [v02-tx I4]. (2) `helix-core` depends only on `rlib` members; the `crate-type` distinction (`staticlib/cdylib`) lives *only* in `helix-ffi`, so the heavy linker artifact is produced once at the apex (§2). (3) `helix-daita` is behind a cargo feature = "daita" (off in Phase-1) so DAITA code is not even compiled into the lean MVP staticlib [doc 01 §9].

1.2 Workspace Cargo.toml (skeleton)

```
# helix-core/Cargo.toml
[workspace]
members = ["crates/*", "bin/*"]
resolver = "2"

[workspace.dependencies]
tokio = { version = "1", features = ["rt-multi-thread",
    "net", "time", "sync", "macros"] }
tokio-util = "0.7" # CancellationToken
bytes = "1"
async-trait = "0.1"
thiserror = "1"
zeroize = "1" # SecretBytes scrubbing (§11.4.10)
```

```

boringtun    = "*"                                # helix-wg backend (version pinned
                                                    at Phase-0 graduation; UNVERIFIED exact)

# frb + UniFFI live ONLY in helix-ffi (§4); not workspace-wide.

[profile.release]                                # the lean default (§3.1) – every
    crate inherits
opt-level    = "z"
lto          = "fat"
codegen-units = 1
panic       = "abort"
strip       = true

```

2. Per-platform library artifacts (how the core ships)

The **only** crate that changes shape per platform is `helix-ffi`; everything below it is `rlib` [`research-flutter_ffi` §3]. Two binding generators read one Rust surface (the canonical Mullvad/WARP pattern [04_ARCH §5.1, `research-flutter_ffi` §2]):

Consumer	Linkage	crate-type	Generator
Dart (all in-process FFI platforms)	linked or dlopened by the Flutter engine	staticlib (iOS/macOS) / cdylib (Android/Linux/Windows)	frb v2 (<code>flutter_rust_bridge_codegen</code> → <code>frb_generated.dart</code> + Rust glue)
iOS / macOS NE (Swift)	linked into the <code>NEPacketTunnelProvider</code> extension	staticlib → xcframework	UniFFI (Swift) + <code>cbindgen</code> for the raw <code>attach_tun</code> C-ABI
Android (Kotlin)	<code>System.loadLibrary</code> in <code>VpnService</code>	cdylib per ABI via <code>cargo-ndk</code>	UniFFI (Kotlin) + JNI for <code>attach_tun(fd)</code>
Windows (C#)	privileged service hosts the core	cdylib	<code>cbindgen</code> C header (named pipe IPC marshals the FFI verbs)
Linux (Dart-FFI / in-process)	<code>DynamicLibrary.open</code>	cdylib	frb v2
HarmonyOS NEXT (ArkTS)	NAPI <code>.so</code> , ArkTS bridge	cdylib	<code>cbindgen</code> / N-API (UniFFI-Kotlin not available) <code>cbindgen</code> C ABI

Consumer	Linkage	crate-type	Generator
Aurora OS (C++/Qt)	linked into the Qt backend	cdylib/ staticlib	

Generator decision (D-CORE-1, surfaced per §11.4.66). frb v2 for Dart, UniFFI for native Swift/Kotlin, cbindgen C-ABI where UniFFI is thin. flutter_rust_bridge 2.12.0 is a Flutter Favorite with first-class StreamSink support — the canonical way to push the status stream Rust→Dart [research-flutter_ffi §1/§7]. uniffi-dart is community WIP and "should not be trusted" [research-flutter_ffi §2], so UniFFI is used **only** for its first-party Kotlin/Swift output where a native extension links the core directly (iOS NE, Android VpnService) [research-flutter_ffi §2/§7]. HarmonyOS/Aurora fall back to cbindgen C-ABI because neither has UniFFI-target support [research-flutter_ffi §4/§5]. **frb-crate version MUST equal the Dart-dep version** — a drift is a build-blocking finding (§12 UT/IT) [research-flutter_ffi §1 caveat].

2.1 Build commands (per target)

```
# Android: cdylib per ABI, bundled into jniLibs [research-
ios_android §3]
cargo ndk -t arm64-v8a -t armeabi-v7a -t x86_64 -o ../apps/access/
android/app/src/main/jniLibs build --release

# iOS/macOS: staticlib → xcframework, nightly build-std for the
smallest NE footprint (§3.2)
cargo +nightly build --release -p helix-ffi --target aarch64-
apple-ios \
-Z build-std=std,panic_abort -Z build-std-
features=panic_immediate_abort,optimize_for_size
# then: lipo/xcodesbuild -create-xcframework

# Linux/Windows desktop: cdylib loaded via Dart FFI DynamicLibrary
cargo build --release -p helix-ffi --target x86_64-unknown-linux-
gnu
```

3. Size / memory build strategy

The iOS Network-Extension memory ceiling is the architecture (F-I6) [04_ARCH §5.6 CI4, research-ios_android §1/§4]. The strategy has two halves: the **build profile** (smaller code → fewer mapped pages → lower resident footprint) and the **runtime budget** (bounded buffers, no per-flow growth).

3.1 Release profile (the lean default, every crate inherits)

```
# [profile.release] from the workspace Cargo.toml (§1.2)
[research-ios_android §2]
opt-level      = "z"      # smallest code. (Evaluate "s": often
                          ~similar size, faster – §3.4 measure)
lto            = "fat"    # whole-program optimization across crates
codegen-units = 1        # better cross-module size opt (slower
                          compile, accepted)
panic         = "abort"  # drops unwinding tables + landing pads →
                          smaller + sometimes faster
strip         = true     # strip symbols (Rust 1.59+), ~3–8% extra
                          reduction
```

panic = "abort" is doubly motivated: it shrinks the binary AND removes unwinding, which the FFI boundary requires anyway — **a Rust panic! must never unwind across the C-ABI into Swift/Kotlin/Dart** (UB) [research-ios_android §2].

Corrected panic-boundary contract (fixes a self-contradiction, §11.4.6). An earlier draft of this section claimed the helix-ffi entry points wrap their bodies in `std::panic::catch_unwind`-equivalent guards that convert a caught panic into `CoreError::Internal`. **This is impossible under panic=abort:** `catch_unwind` requires the *unwind* panic strategy to have anything to catch; under abort, the process terminates at the panic site before any unwinding (and therefore any catching) can occur. The two cannot coexist in one profile — the corrected, honest contract is:

1. **No panic is ever caught at the FFI boundary.** With `panic=abort` set workspace-wide (§1.2), a panic! anywhere in helix-core/helix-ffi terminates the **entire hosting process** — the iOS NE extension, the Android app process, the Windows service, or the edge binary. This is a real availability risk (an aborted NE extension is not guaranteed a prompt OS-triggered relaunch), not a caught-and-recovered error.
2. **The mitigation is defensive-coding rigor at the boundary, not runtime recovery.** Every helix-ffi entry point (§4.1) MUST validate externally-supplied input and return a typed `CoreError` variant (`Config/BadFd/Auth/...`) instead of ever reaching a `.unwrap()`/`.expect()`/`panicking-index/unreachable!()` on FFI-boundary data. A `CoreError` is the ONLY sanctioned way an FFI call reports failure — a panic is by definition a bug, not an expected error path.
3. **Enforced mechanically, not by convention.** helix-ffi (and any crate an FFI entry point calls into on the hot path) carries a `clippy.toml`/lint config denying `unwrap_used`, `expect_used`, `panic`, `unimplemented`, `todo`, and `indexing_slicing` — a pre-build gate (§11.4.4(b) layer 1),

with a paired §1.1 mutation (reintroduce a denied lint call → the gate FAILs).

4. **Accepted tradeoff, stated honestly.** A genuine internal-invariant violation (a real bug, not user error) aborts the host process. This is the deliberate, accepted cost of the panic=abort size optimization (this section); the alternative (panic=unwind + a real catch_unwind) would re-admit the unwinding machinery §3.2's build-std reduction removes, undermining the iOS memory budget (§3.3) this whole strategy exists to protect. The lint-enforced no-panic-on-boundary-data discipline (point 3) is what keeps this an acceptable risk rather than a routine crash surface.

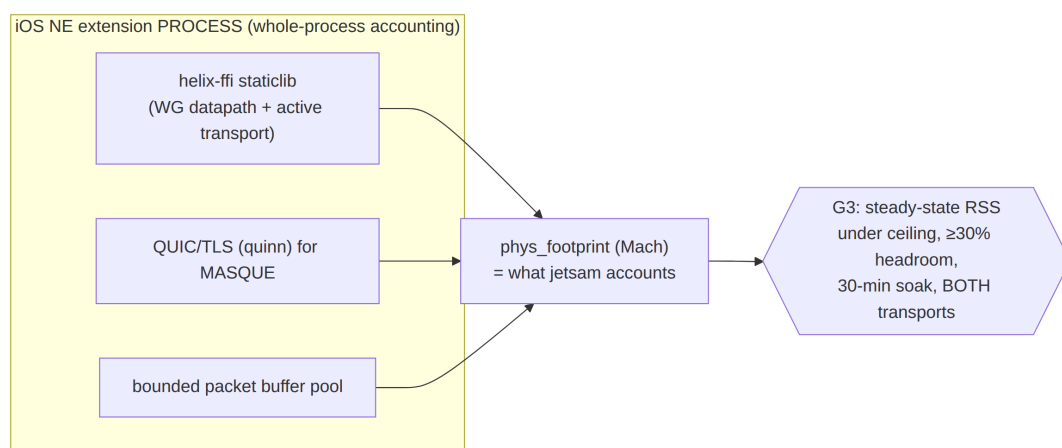
3.2 iOS-only: build-std for the largest reduction

For the aarch64-apple-ios target the standard library is rebuilt with size features — this drops the gimli backtrace machinery and all panic-formatting code from std [research-ios_android §2]:

```
-Z build-std=std,panic_abort
-Z build-std-features=panic_immediate_abort,optimize_for_size
```

Representative published reduction: ~43 % over the baseline release profile [research-ios_android §2, markaicode]. This is the make-or-break lever for the G3 memory gate (§3.3). **staticlib (not dylib) for iOS** — no dynamic-loading entitlement friction, and the linker dead-strips unreferenced symbols [research-ios_android §2].

3.3 Runtime memory budget & the G3 gate



Fact	Value	Source
Documented NE limit (iOS 15–18)	50 MiB (whole process incl. all linked libs)	[research-ios_android §1, Apple DTS]

Fact	Value	Source
Observed real-world jetsam kill	~15 MB on iOS 17.3.1 (root cause PENDING_FORENSICS)	[research-ios_android §1]
Dominant kill cause	memory spikes during upload-heavy throughput, not steady-state	[research-ios_android §1, sing-box #3976]
Engineering target (this spec)	steady-state ≤ ~12-15 MB working set , never plan to 50 MiB	[research-ios_android §1/§4]
The limit	undocumented, MUST NOT be hardcoded — measure on real devices	[research-ios_android §1, Quinn DTS]

Runtime rules: (1) **bounded packet-buffer pool** sized to negotiated MTU + headroom (§6.3), never an arbitrary large read buffer [research-ios_android §1]; (2) **WG-class low-state datapath** — no TCP connection tracking, minimal persistent heap [research-ios_android §1/§4]; (3) **measure phys_footprint / task_vm_info** (the value jetsam accounts against — UNVERIFIED exact Mach field, verify on device) sampled from inside the extension, plus Instruments Allocations/VM-Tracker [research-ios_android §1].

G3 (make-or-break) gate [04_P0 §6, doc 03 §5.1]: build the same helix-ffi as aarch64-apple-ios with §3.1+§3.2; drive a **1 GB transfer on a real device** (Simulator memory is *not* representative); sample the **extension process** RSS; run **plain-UDP and MASQUE separately** (QUIC buffers cost more). **Pass = steady-state under the device-enforced ceiling with ≥ 30 % headroom across a 30-minute soak, both transports.**

Documented fallbacks in order (each a real product decision — D-CLIENT-1, surfaced [04_P0 §6.4, doc 03 §14]): (a) shrink buffers / cap QUIC flow-control windows in the iOS build; (b) move MASQUE off-device for iOS (plain WG + on-path obfuscation only — partial feature loss); (c) split the core so only the lean WG datapath is in-extension and QUIC negotiation lives in the app via app-extension IPC.

3.4 Android memory posture (contrast)

Android VpnService runs as a normal foreground service — **no hard 15/50 MiB jetsam cap**; standard per-app lowmemorykiller applies, far more generous [research-ios_android §3]. The severe constraint is **iOS-only**; the same lean build is used everywhere (smaller is always better), but the budget is gated on iOS.

3.5 App-size budgets (the install-size half)

Budget	Target	Means	Source
Mobile install (per ABI)	Access < ~ 15-20 MB	--split-per-abi, lean staticlib (LTO+strip), tree-shake	[04_UI §10]
Core staticlib size	record per target; smallest is the goal	§3.1/§3.2; audit fat deps (regex/formatting/async) [research-ios_android §2 bitdrift]	[research-ios_android §2]

4. The helix-ffi crate — the surface beneath the FFI

helix-ffi/src/api.rs is the **only hand-authored FFI source**; both generators read it [04_UI §5, doc 03 §3.1]. It is a thin, panic-safe adapter over the helix-core Orchestrator (§6) — it holds the live Orchestrator behind a process-global handle and translates FFI calls into orchestrator methods. The surface is **frozen at the Phase-0 → Phase-1 graduation** [04_P0 §11]; its shape matches [04_UI §5 / doc 03 §3] and its TunnelStatus matches [v02-orch §4.1] via the §5 projection.

4.1 api.rs — the exported function surface

```
// helix-ffi/src/api.rs – frb v2 + UniFFI generate Dart/Swift/
// Kotlin/C bindings.
// Transport/WG/orchestrator internals are v02/doc-01's; this is
// the EXPORTED surface only.
use flutter_rust_bridge::frb;

pub struct ClientConfig {
    pub map_path_or_session:
        String, // Phase 0: path to map.json; Phase 1+: control-
               // plane session token
    pub transport: String, // "auto" | "plain" |
                        "masque" | "shadowsocks" | "uot" | "lwo"
    pub mode: CoreMode, // Client | Connector (F-I3)
}

#[frb(mirror(CoreMode))]
pub enum CoreMode { Client, Connector }

// ---- lifecycle (LOGIC ONLY – NOT the OS tunnel; the shim owns
// that, F-I2) ----
/// Build the Orchestrator (§6.1), arm the kill-switch closed
// (§8.2), spawn the
```

```

/// loops, emit Connecting. Idempotent: a second start while
/// running is Err(AlreadyStarted) (§6.1).
pub async fn start(cfg: ClientConfig) -> Result<(), CoreError>;
/// Graceful stop: cancel loops, FIN transport, revert kill-
/// switch+DNS per §8.6, emit Down{stopped}.
pub async fn stop() -> Result<(), CoreError>;

// ---- the live status stream (maps to the orchestrator
// broadcast, §5) ----
/// frb StreamSink → Dart broadcast Stream. Many concurrent sinks
/// are allowed (broadcast
/// fan-out, §5.3); each call subscribes a fresh receiver and
/// immediately re-emits the
/// cached latest status (§5.2). Returns Err(NotStarted) if the
/// core is not running.
pub fn status_stream(sink: StreamSink<FfiTunnelStatus>) ->
    Result<(), CoreError>;

// ---- exits / routing (Phase 1+) ----
pub async fn exits() -> Result<Vec<ExitOption>, CoreError>;
pub async fn set_exit(id: String, multi_hop_chain:
    Option<Vec<String>>) -> Result<(), CoreError>;

// ---- privacy shields (pure logic – no OS tunnel touched, F-I2)
// ----
pub async fn set_shields(s: Shields) -> Result<(), CoreError>;

// ---- connector mode only (§9) ----
pub async fn advertise(cidrs: Vec<String>) ->
    Result<AdvertiseResult, CoreError>;

// ---- desired-state push (Phase 1: bridges WatchNetworkMap;
// Phase 0: file-watch) (§7) ----
pub async fn apply_map(map_json: String) -> Result<MapApplied,
    CoreError>;

// ---- shim handoff: the core NEVER opens the TUN; the shim hands
// it a packet fd / pump (F-I2, §6) ----
pub fn attach_tun(fd: i32) -> Result<(), CoreError>; // Android
// ParcelFileDescriptor / Linux tun fd
pub fn detach_tun() -> Result<(), CoreError>;

```

Canonical FFI contract (R7-reconciled). Every FFI verb returns the typed `CoreError` (§10.1), never `anyhow::Result` — `anyhow` stays *internal* to the orchestrator and is converted at the boundary via `From` impls (see `v04-client/ffi-surface.md` §8.1, the canonical owner of the exported surface). The Dart-facing status enum is `FfiTunnelStatus` (§5.1); `ffi-surface.md` writes it path-qualified as `ffi::TunnelStatus` purely to contrast with the orchestrator's `core::TunnelStatus` — it is **the same Rust type**.

4.2 Mirrored value types (frb #[frb(mirror)] → identical Dart/Swift/Kotlin)

```
#[frb(mirror(Shields))]  
pub struct Shields {  
  pub kill_switch: bool,           // → KillSwitchConfig.mode  
    Strict/Off (§8.2)  
  pub dns_protection: bool,        // →  
    DnsConfig.block_plaintext_53 (§8.3)  
  pub daita: bool,                 // feature-gated; Phase 2  
    [doc 01 §9]  
  pub post_quantum: bool,          // PQ PSK; Phase 2 [SYN §4]  
  pub split_tunnel: Vec<String>,   // per-route bypass; per-APP  
    split handled in the shim layer  
}  
  
#[frb(mirror(ExitOption))]  
pub struct ExitOption {  
  pub id: String, pub kind: String, // "privacy_exit" |  
    "network"  
  pub label: String, pub country: Option<String>, pub rtt_ms:  
    Option<u32>,  
  pub jurisdiction: Option<String>, // for multi-hop  
    labels  
}  
  
#[frb(mirror(AdvertiseResult))]  
pub struct AdvertiseResult { pub accepted: Vec<String>, pub  
  conflicts: Vec<String> }  
  
#[frb(mirror(MapApplied))]  
pub struct MapApplied { // §7 reconcile  
  outcome surfaced to the UI  
  pub peers_added: u32, pub peers_removed: u32, pub  
    peers_changed: u32,  
  pub dns_changed: bool, pub restart_free: bool, // always  
    true (0-I12); a false here is a bug  
}
```

4.3 Panic-safety & the process-global handle

```
// helix-ffi/src/state.rs – exactly one live Orchestrator per  
  process.  
use once_cell::sync::OnceCell;  
use tokio::sync::Mutex;  
static RUNTIME: OnceCell<tokio::runtime::Runtime> =  
  OnceCell::new(); // multi-thread runtime  
static ORCH: OnceCell<Mutex<Option<helix_core::Orchestrator>>> =  
  OnceCell::new();  
  
// Every async fn body is dispatched on RUNTIME and guarded:  
// - frb v2 marshals async Rust to a Dart Future; a Rust Err  
  becomes a Dart exception.
```

```
// - with panic=abort (§3.1) a genuine bug ABORTS the process –
//   it is NEVER caught
//   at this boundary (no catch_unwind is possible under abort;
//   see §3.1's corrected
//   panic-boundary contract). "Guarded" here means INPUT-
//   VALIDATED, not panic-caught:
//   every body validates its arguments and returns CoreError on
//   anything unexpected,
//   so the only path left to an abort is a genuine internal-
//   invariant bug, not bad input.
// - CoreError (§10) is the structured, surfaced failure; never
//   a silent bluff (§11.4.6).
```

UNVERIFIED: whether frb v2's generated async runtime obviates a hand-rolled RUNTIME (frb manages a worker pool) — confirm at codegen; if frb owns the runtime, RUNTIME collapses to frb's [research-flutter_ffi §1]. The single-live- orchestrator invariant (one TUN, one tunnel per app process) is enforced regardless.

5. TunnelStatus — core enum ⇌ FFI projection

The **core** broadcast type is the frozen 5-variant enum owned by [v02-orch §4.1]:

```
// helix-core/src/status.rs (OWNED by v02-orch §4.1 – reproduced
//   for the seam, MUST match)
#[derive(Clone, Debug, PartialEq, Eq)]
pub enum TunnelStatus {
    Connecting, // dialling the
                first/seeded rung
    Handshaking, // carrier up,
                WG handshake in flight
    Connected { transport: String, rtt_ms: u32 }, // e.g.
                ("masque-h3", 23)
    Reconnecting, // a working
                tunnel dropped; re-dialling
    Down { reason: String }, // terminal for
                this attempt (§10 reasons)
}
```

The **FFI-facing** enum (the Dart/Swift/Kotlin surface, doc 03 §3.1) is a strict *projection* of the core enum plus three FFI-only additions. The projection is the **only** place the two differ; the §12 contract test asserts the mapping byte-for-byte so they can never silently drift.

```
// helix-ffi/src/api.rs
#[frb(mirror(FfiTunnelStatus))]
pub enum FfiTunnelStatus {
    Disconnected, //
                FFI-only: clean idle (pre-start / post user-stop)
```

```

    Connecting, //
    ← core Connecting

    Handshaking, //
    ← core Handshaking
    Connected { transport: String, path: String, rtt_ms:
    u32 }, // ← core Connected; `path`="direct"|"relay" (§5.1)

    Reconnecting, //
    ← core Reconnecting
    Down { reason: String }, // ←
    core Down{reason} (unexpected drop)
    Danger { kind: String }, //
    FFI-only: "leak" | "killswitch_tripped" (§5.2)
}

```

5.1 The projection map (core → FFI)

Core TunnelStatus	FFI FfiTunnelStatus	Notes
(no live orchestrator)	Disconnected	pre-start() and after stop() → Down{stopped} is re-mapped to Disconnected for the UI (clean idle)
Connecting	Connecting	1:1
Handshaking	Handshaking	1:1
Connected{transport, rtt_ms}	Connected{transport, path, rtt_ms}	path derived: "relay" if the active leg is DERP-style relay, else "direct" (P2P/NAT-traversal — doc 01 §8); Phase-1 default "direct"
Reconnecting	Reconnecting	1:1
Down{reason: "stopped"}	Disconnected	user-stop is clean, not a danger
Down{reason: r} (r≠stopped)	Down{reason: r}	unexpected/terminal drop (kill-

Core TunnelStatus	FFI FfiTunnelStatus	Notes
		switch stays closed, §8)
(kill-switch trip / leak detected)	Danger{kind}	FFI-only overlay derived from the §8 sub-machine + leak probe; drives the red stateDanger palette (doc 03 §7)

Consistency note (§11.4.6). The core never emits Disconnected or Danger — they are FFI-layer derivations, so the v02 5-variant broadcast contract is untouched. Danger is **not** a guessed UI state: it is raised only on a concrete signal — the §8 kill-switch sub-machine reporting killswitch_tripped, or an explicit DNS/leak probe firing — never inferred from prose.

5.2 The StreamSink pump

```
// helix-ffi/src/api.rs – status_stream wiring
pub fn status_stream(sink: StreamSink<FfiTunnelStatus>) {
  let mut rx = orchestrator().subscribe(); //
  broadcast::Receiver<core::TunnelStatus> [v02-orch §2.1]
  spawn_on_runtime(async move {
    // emit initial Disconnected so the UI has a value before
    // the first transition (CI2)
    let _ = sink.add(FfiTunnelStatus::Disconnected);
    loop {
      match rx.recv().await {
        Ok(core_s) => { let _ =
          sink.add(project(core_s)); } // §5.1 map
        Err(broadcast::error::RecvError::Lagged(_)) =>
          continue, // skip-to-latest, NOT an error [v02-orch
                    §4.6]
        Err(broadcast::error::RecvError::Closed) =>
          break, // orchestrator gone → stream ends
      }
    }
  });
}
```

frb v2 turns this StreamSink<FfiTunnelStatus> into a Dart Stream that feeds a Riverpod StreamProvider.autoDispose [research-flutter_ffi §6]. **Riverpod 3.0 pauses the subscription when no widget listens** — so the FFI pump is torn down when the UI backgrounds

the status, and re-subscribes on resume; the core keeps running regardless (the broadcast sender outlives any receiver) [research-flutter_ffi §6].

5.3 Receiver contract (Dart side, doc 03 §8.2 honors this)

Lagged MUST be treated as "re-read latest", never an error; Closed ends the stream. The status is edge-coalesced (emissions are rare — state changes), so the 64-slot broadcast ring [v02-orch §4.2] never lags unless a receiver is wedged.

5.4 No poll getter (F-I1)

The FFI exposes **no** `is_connected()` / `current_status()` synchronous getter. The UI's only source of protection truth is `status_stream()` — this structurally prevents the "UI says connected while the tunnel is down" bug (CI2) [04_UI §4.2].

5.5 What the FFI MUST NOT return (F-I5)

No endpoint, SSID, gateway IP, per-packet/per-flow datum, or SecretBytes ever crosses the FFI. `Connected.transport` is a *kind* label; `rtt_ms` is an aggregate EWMA [v02-orch §4.5, v02-tx I5]. A code-review gate (§11.4.142) rejects any FFI return embedding an endpoint or secret.

6. FFI lifecycle & the TUN-fd handoff (the shim seam)

The seam that prevents the classic VPN bug [doc 03 §3.3, v02-orch §2.1]: **lifecycle commands flow UI → shim → core** (the OS owns the tunnel process), while **status events flow core → FFI → UI** (the core owns protection truth). `set_shields/set_exit/exits` are pure logic and go straight UI → FFI without touching the OS tunnel (F-I2).

6.1 `start()` → builds the Orchestrator

```
// helix-ffi/src/api.rs (body sketch) – drives helix-core; v02-orch §2.1 owns Orchestrator::start
pub async fn start(cfg: ClientConfig) -> Result<(), CoreError> {
    let mut guard = orch_cell().lock().await;
    if guard.is_some() { return Err(CoreError::AlreadyStarted); } // idempotent: no double tunnel
```

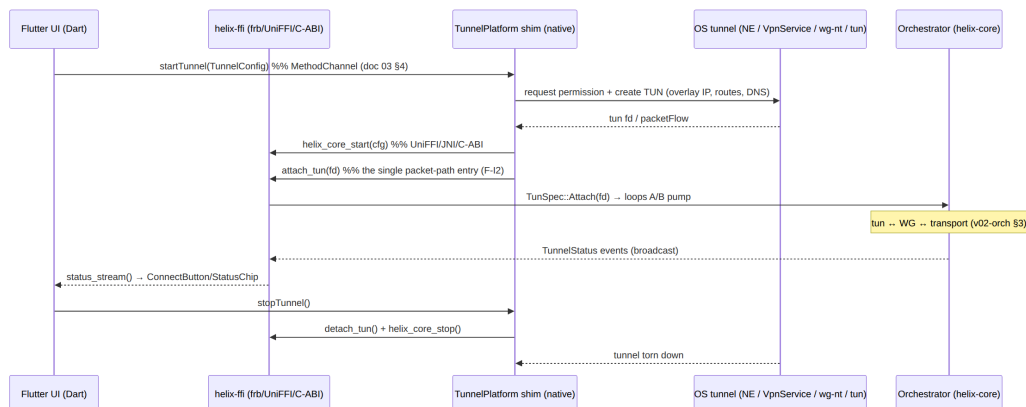
```

let initial_map =
  load_map(&cfg.map_path_or_session).await?; // Phase 0:
  file; Phase 1: session→WatchNetworkMap
let oc = helix_core::OrchestratorConfig {
  mode:
    cfg.mode.into(), // F-
    I3
  tun:
    TunSpec::AttachLater, // the
    shim calls attach_tun (§6.2) – core does NOT open it
    initial_map,
    transport_policy:
    TransportPolicy::from_hint(&cfg.transport),
    kill_switch:
    KillSwitchConfig::default(), // armed-
    closed-until-connected (§8.2)
    dns: DnsConfig::from(&initial_map),
    tunables:
    CoreTunables::default(), // §11
};
*guard =
  Some(helix_core::Orchestrator::start(oc).await?); //
  arms kill-switch CLOSED, spawns loops, emits Connecting
Ok(())
}

```

start() does **not** open the TUN. The orchestrator is built in TunSpec::AttachLater and idles loop A/B until attach_tun(fd) injects the packet pipe (F-I2). This is the contract that keeps the *only* per-platform code in the shim [04_UI §5/§6].

6.2 The TUN-fd handoff (the C-ABI seam, per OS)



Per-OS fd source [research-ios_android §3, doc 03 §5]:

- **Android (Kotlin/JNI):** ParcelFileDescriptor.detachFd() transfers ownership; the int fd is passed into JNI; the core read(2)/write(2)s directly on it. Every outbound transport socket the core opens MUST be protect()ed (ideally native

protectFromVpn to avoid per-socket JNI upcalls) so the tunnel's own packets don't loop back into the tun [research-ios_android §3].

```
// apps/access/android/.../HelixVpnService.kt
external fun coreStart(sessionToken: String): Int //
    JNI → helix_ffi start()
external fun attachTun(fd: Int): Int //
    JNI → helix_ffi attach_tun()
// ...
val pfd: ParcelFileDescriptor = builder.establish()!! //
    OS creates the TUN
coreStart(cfg.sessionToken); attachTun(pfd.detachFd()) //
    hand raw fd to the core
```

- **iOS/macOS (Swift, NE):** the extension owns packetFlow; there is no integer fd — instead the shim wires the read/write callbacks. The C-ABI exposes a tun_out push and a set_tun_writer callback (the core has no fd, it pumps via the closures):

```
// apps/access/ios/HelixTunnel/PacketTunnelProvider.swift
helix_core_start(cfg.sessionToken) //
    UniFFI/cbindgen
packetFlow.readPackets { packets, _ in
    for p in packets { helix_core_tun_out(p) } //
        core encrypts + sends via transport
}
helix_core_set_tun_writer { data in //
    core → OS: decrypted IP packets
    self.packetFlow.writePackets([data], withProtocols:
        [AF_INET])
}
```

- **Linux (in-process):** the core opens/owns the tun fd directly (a setcap cap_net_admin helper or polkit-gated daemon covers privilege); the Phase-0 G5 reference platform "the core can drive the TUN directly" [04_P0 §9, doc 03 §5.4].
- **Windows:** a privileged service hosts the core + the wireguard-nt/wintun adapter; the Flutter app marshals the FFI verbs over a named pipe; the wintun ring is the core↔OS packet path [doc 03 §5.3].
- **HarmonyOS NEXT (ArkTS):** the VpnExtensionAbility tun is bridged via NAPI to the .so; honest SKIP-with-reason (§11.4.3) where the SIG port lacks the VPN API [research-flutter_ffi §5, research-ios_android n/a].
- **Aurora OS (Qt/C++):** the Qt backend owns the tun; the C-ABI receives the fd [doc 03 §5.6].

6.3 The C-ABI surface for the native packet path (cbindgen)

```
// helix_ffi.h (cbindgen-generated) – the raw seam UniFFI does not
// cover (packet hot path)
int32_t helix_core_start(const char* session_token, int32_t
mode); // 0=Client 1=Connector
int32_t helix_core_attach_tun(int32_t
fd); // Android/Linux fd path
void helix_core_tun_out(const uint8_t* pkt, uintptr_t
len); // iOS push-in path (one IP packet)
void helix_core_set_tun_writer(void (*writer)(const uint8_t*,
uintptr_t)); // core → OS callback
int32_t helix_core_detach_tun(void);
int32_t helix_core_stop(void);
```

The read buffer the core allocates for `tun_out`/the fd read loop is **bounded to the negotiated MTU + headroom** (§3.3), never an arbitrary large value — the single strongest lever against iOS jetsam spikes [research-ios_android §1].

7. The network-map reconciler beneath the FFI

Ownership. The reconciler's diff/apply bodies are specified in depth by [v02-orch §6]. This section specifies the **FFI-driven entry** (`apply_map`), the consumed RouteMap/MapDelta shapes, and the *client-core* obligation to bring WG peers up/down and update AllowedIPs restart-free — kept byte-consistent with [v02-orch §6.2/§6.3].

The core is a **declarative reconciler** [04_ARCH §4.4, 04_P0 §10]: it diffs a *desired* RouteMap (Phase 1: pushed by WatchNetworkMap, doc 02; Phase 0: a file-watch on `map.json`) against live state and converges — peers up/down, AllowedIPs, transport policy, DNS — **without restarting** (O-I12), < 1 s convergence [04_ARCH §4.4].

7.1 Consumed shapes (frozen, from [v02-orch §6.2])

```
// helix-route (consumed by the reconciler) – MUST match v02-orch
// §6.2 byte-for-byte
pub struct RouteMap {
  pub self_overlay:
    IpAddr, // this node's overlay IP (ULA /48,
    D4)
  pub peers: Vec<PeerRoute>, // already policy-
    filtered (need-to-know, I6)
  pub dns: Vec<IpAddr>, // tunnel DNS servers
    (§8.3)
```

```

    pub transport_policy: TransportPolicy, // per-leg ladder
      order/pin/budget [v02-tx §6.2]
    pub daita: Option<DaitaMachines>,      // optional shaping
      config (data, not code; feature-gated)
  }
  pub struct PeerRoute {
    pub wg_pubkey: [u8; 32],
    pub endpoint_candidates: Vec<SocketAddr>, // NAT traversal
      (Phase 2)
    pub allowed_ips: Vec<IpNet>,             // overlay prefixes
      this peer is next hop for
    pub via_connector: Option<SiteId>,        // 4via6 site
      disambiguation [doc 01 §6.1]
  }
  pub struct MapDelta {                    // produced by
    diff(), surfaced to the FFI as MapApplied (§4.2)
    pub peers_added: Vec<PeerRoute>,
    pub peers_removed: Vec<[u8; 32]>,        // pubkeys to tear
      down (revocation path)
    pub peers_changed: Vec<PeerRoute>,       // allowed_ips /
      endpoint changed
    pub dns_changed: bool, pub policy_changed: bool, pub
      transport_policy_changed: bool,
  }
}

```

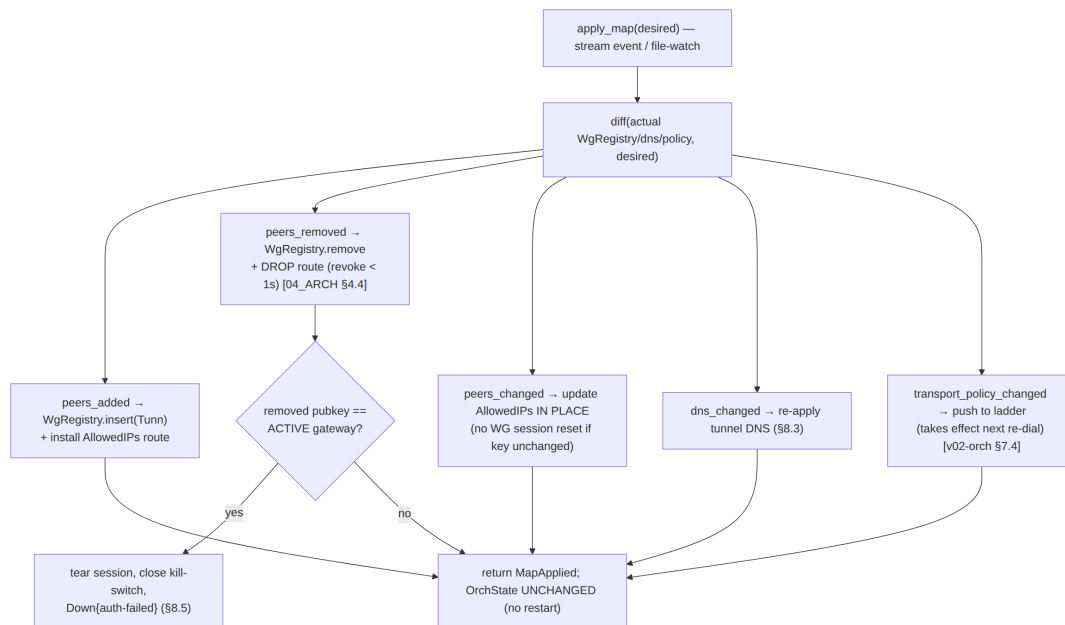
7.2 apply_map — the FFI-driven converge

```

// helix-ffi/src/api.rs – Phase 1 wires WatchNetworkMap → this;
// Phase 0 the file-watch task calls it.
pub async fn apply_map(map_json: String) -> Result<MapApplied,
  CoreError> {
  let desired: RouteMap =
    serde_json::from_str(&map_json)?; // Phase-0 stand-
    in shape == Phase-1 stream
  let delta =
    orchestrator().apply_map(desired).await?; // v02-
    orch §6.4 in-place converge (0-I12)
  Ok(MapApplied {
    peers_added: delta.peers_added.len() as u32,
    peers_removed: delta.peers_removed.len() as u32,
    peers_changed: delta.peers_changed.len() as u32,
    dns_changed: delta.dns_changed,
    restart_free:
      true, // ALWAYS
      true; a false here is a bug (0-I12)
  })
}

```

7.3 Apply flow (restart-free) — what the client core does per delta class



- **peers_added** → insert a Tunn into the WG registry and install its allowed_ips as overlay routes; the new peer is reachable without a tunnel restart [v02-orch §6.4].
- **peers_removed** → remove the Tunn and DROP its routes; if the removed pubkey is the **active** gateway (revocation-of-self), tear the session, close the kill-switch, emit Down{auth-failed} → FFI Down{"auth-failed"} (§8.5). Effective < 1 s [04_ARCH §4.4].
- **peers_changed** touching only allowed_ips → update the routing map **in place**, **no WG session reset** (the key is unchanged) — the "policy edit < 1 s, no restart" MVP gate (AC5) [v02-orch §6.4, 04_P1 §11.2].

7.4 Reconcile is a side task (fail-static, I3)

The reconcile source dying (stream disconnect / file-watch error) never tears the tunnel — the loops keep forwarding on the last-applied map (fail-static) [04_ARCH §2.1, v02-orch §6.5]. The FFI surface is unaffected; apply_map simply isn't called until the source recovers (Phase-1 stream re-subscribes with its own backoff, independent of the data-plane backoff).

8. Kill-switch + DNS state machine

Ownership. The kill-switch/DNS coupling bodies are specified by [v02-orch §8]. This section specifies how the **core configures and exposes** them beneath the FFI (the Shields mapping, the Danger projection) — kept consistent with [v02-orch §8.4].

The kill-switch and DNS-leak protection are **owned by the core state machine**, never hand-edited firewall rules [04_ARCH §5.6/§7]. Every open/close and DNS apply/revert is a side effect of an OrchState transition (O-I10), so there is no window where the UI shows "connected" but the firewall is open (F-I4).

8.1 Config (from Shields + RouteMap)

```
// helix-core (configured by start() from ClientConfig+RouteMap;
// v02-orch §8.2/§8.3 owns bodies)
pub struct KillSwitchConfig {
    pub mode: KillSwitchMode,           // Strict |
        Permissive | Off
    pub allow_lan: bool,                 // permit RFC1918
        LAN while closed (printers etc.)
    pub allow_gateway_endpoints: Vec<IpNet>, // gateway IPs the
        transport MUST reach (stay open)
}
pub enum KillSwitchMode { Strict, Permissive, Off }
pub struct DnsConfig {
    pub servers: Vec<IpAddr>,           // tunnel DNS (from
        RouteMap.dns)
    pub block_plaintext_53: bool,        // drop off-tunnel
        :53 when Connected (leak guard)
    pub apply: DnsApplyMethod,          // resolvconf |
        systemd-resolved | /etc/resolv.conf | shim
}
```

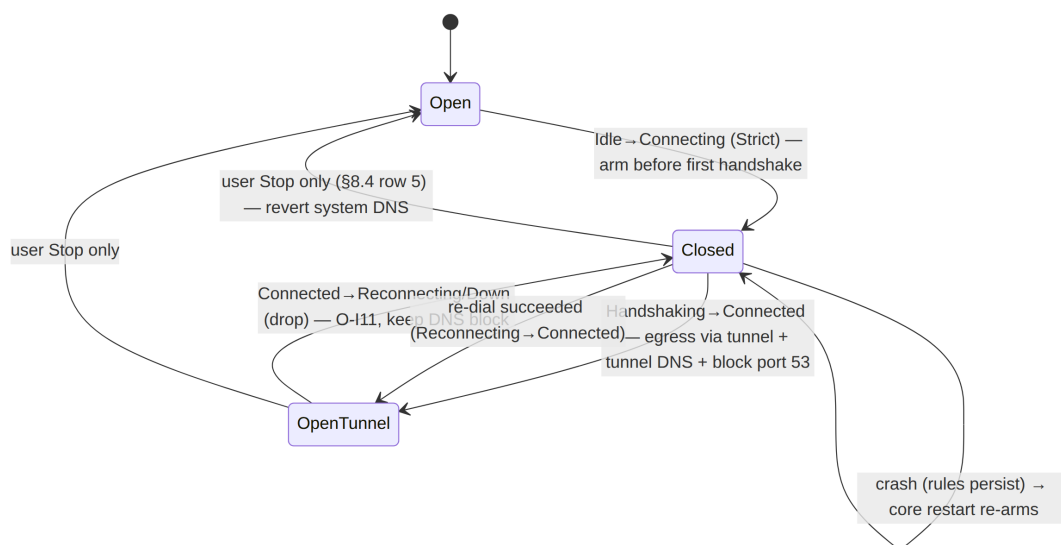
The FFI Shields (§4.2) maps onto these: kill_switch=true ⇒ KillSwitchMode::Strict; dns_protection=true ⇒ block_plaintext_53=true. set_shields() re-applies live (no restart) [doc 03 §3.1].

8.2 The side-effect matrix (from [v02-orch §8.4], drives Danger projection)

Transition	Kill-switch	DNS	FFI projection
Idle → Connecting (Strict)	install closed (allow gw	system DNS	Connecting

Transition	Kill-switch	DNS	FFI projection
	endpoints only)		
Handshaking → Connected	open (egress via tunnel)	apply tunnel DNS + block off-tunnel :53	Connected
Connected → Reconnecting (drop)	close (O-I11)	keep tunnel-DNS block (fail-closed)	Reconnecting
* → Down{reason≠stopped}	stay closed	keep block	Down{reason} (or Danger{killswitch_tripped} if a leak probe fired)
ShuttingDown → Down{stopped} (user)	revert to off (open)	revert system DNS	Disconnected
crash / panic	closed rules remain	block remains	(process restarts; start() re-arms)

8.3 Kill-switch + DNS sub-machine (from [v02-orch §8.7])



8.4 Danger projection (the FFI-only overlay, §5.2)

The FFI raises `Danger{kind}` when, and only when, a concrete signal fires:

- `kind="killswitch_tripped"` — the sub-machine reports the kill-switch closed *while the user intends connected* (a drop the user can see); or
- `kind="leak"` — an explicit DNS/leak probe detects off-tunnel egress.

Danger overrides any optimistic UI intent and paints the red stateDanger palette immediately (doc 03 §7/§8.4). It is never guessed (§11.4.6) — absent a signal the FFI emits Reconnecting/Down, not Danger.

8.5 auth-failed is closed-and-terminal

A revoked device (`peers_removed` includes the active gateway, §7.3) reaches `Down{auth-failed}` with the kill-switch **closed** and DNS **blocked** — a revoked client cannot leak onto the underlay by virtue of being revoked [v02-orch §8.5]. Recovery requires explicit user re-enroll through `start()`.

9. Connector mode (--mode=connector)

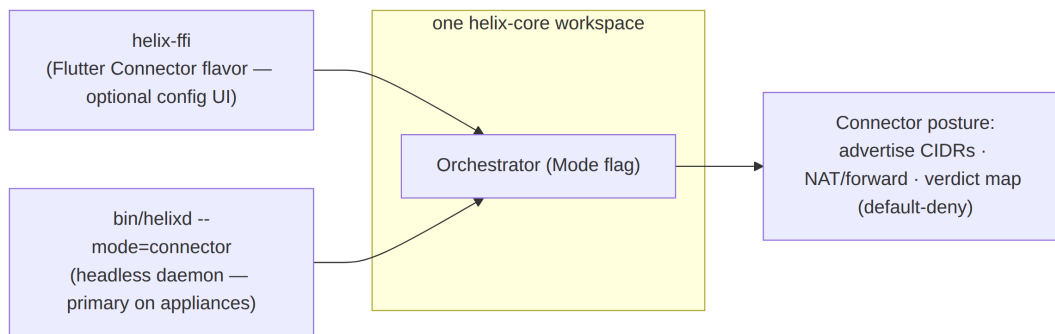
Ownership. The connector forwarding/NAT/verdict-map bodies are [v02-orch §9]. This section specifies the **packaging** — one binary, the mode flag, the headless daemon, and the FFI advertise verb sharing the same core.

9.1 One binary, one core, two postures (F-I3)

The connector is the **same** helix-core Orchestrator with `Mode::Connector` [v02-orch §9.1, 04_P0 §4.5]. The three loops, status stream, reconciler, backoff, and transport layer are byte-for-byte identical to the client; only the routing posture differs (no default-route capture; instead advertise served CIDRs + NAT/forward decapsulated packets into the LAN) [v02-orch §9.1/§9.2]. The connector is **outbound-only** — it dials the gateway, never listens (the founding no-port-forward constraint [SYN §1]).

client: [app traffic] → TUN → WG encrypt → [DAITA] → Transport
→ gateway
connector: gateway → Transport → WG decrypt → forward into
served CIDR (and reverse)

9.2 The two entrypoints share the core



- **Headless daemon** (`bin/helixd --mode=connector`) is the **primary** form on network appliances — no Flutter, no FFI weight, just the core [04_ARCH §5.4, doc 03 §6]. It reads its RouteMap from the control plane (or `map.json`) and runs the same reconciler (§7) + kill-switch knobs (typically `KillSwitchMode::Off` — a connector is an appliance, not a privacy client [v02-orch §9.4]).
- **Flutter Connector flavor** is the **optional** config surface; it links `helix-ffi` and drives the *same* core with `CoreMode::Connector`, calling `advertise(cidrs)` for the advertise UI [doc 03 §6, §3.1].

9.3 advertise and local_denylist (FFI verbs, connector only)

```
// helix-ffi/src/api.rs (connector mode)
pub async fn advertise(cidrs: Vec<String>) ->
    Result<AdvertiseResult, CoreError>;
// → orchestrator pushes the served CIDRs into its RouteMap; the
// gateway reconciles them.
// AdvertiseResult.conflicts surfaces overlapping-CIDR rejections
// to the Connector UI (doc 03 §6).

pub async fn advertise_with_local_denylist(
    cidrs: Vec<String>,
    local_denylist: Vec<String>,
) -> Result<AdvertiseResult, CoreError>;
// → advertises served CIDRs AND the connector's local_denylist to
// the control plane.
// The denylist is sent alongside CIDR advertisements (e.g. in
// the `connector.local_denylist.changed`
// event payload or equivalent agent RPC) so the coordinator/
// edge applies the same precedence rule.
```

The connector accepts a `local_denylist` config value (daemon flag `--local-denylist <cidrs>`, JSON array, or FFI parameter) scoped to its own network. The same precedence rule is enforced both in the control-plane compiler (`svc-policy.md` §4.2) and at the connector's local edge:

1. **Local-deny overrides central-allow** — a connector may tighten policy for its own network.
2. **Central-deny overrides local-allow** — the tenant-wide default-deny/fail-closed invariant wins; a connector cannot use `local_denylist` to re-open a flow the central policy did not grant.
3. The resulting AllowedIPs + edge verdict map equal the **union of central policy minus local-deny** for that connector's traffic.

The connector advertises the `local_denylist` to the coordinator alongside its CIDR advertisements so the edge enforces it consistently; the local edge also applies it as a belt-and-suspenders check in case the control-plane delta is delayed (fail-static, never fail-open).

9.4 Status semantics on the connector

The connector emits the **same** `TunnelStatus` stream — its tunnel to *the gateway* is what `Connected{transport, rtt}` describes; `Down` means "the connector lost its gateway tunnel" [v02-orch §9.4]. The Connector Flutter flavor subscribes to `status_stream()` exactly as Access does (the FFI/UI code is shared, F-I3).

10. Error handling & edge cases

10.1 FFI error taxonomy

```
// helix-ffi/src/error.rs – what crosses the FFI as a Dart
// exception / Swift-Kotlin throw
#[derive(thiserror::Error, Debug)]
pub enum CoreError
{
    // canonical = ffi-surface.md §8.1
    #[error("core not started")]
    NotStarted, // status_stream/stop/apply_map/
    // exits before start
    #[error("core already started")]
    AlreadyStarted, // start() while running (§6.1)
    #[error("invalid config: {0}")]
    Config(String), // bad transport/map path; apply_map
    // bad JSON (§7.2)
    #[error("auth failed (revoked)")]
    Auth, // maps from Down{auth-failed} (§8)
```

```

    #[error("host fatal: {0}")]
    HostFatal(String),    // TUN open / firewall apply failed
                           (§8)
    #[error("tun fd invalid")]
    BadFd,                // attach_tun on a dead/foreign fd
    #[error("internal: {0}")]
    Internal(String),     // panic guard (§4.3) – bug,
                           surfaced not hidden; never a secret (§11.4.10)
}
// The orchestrator's internal anyhow / OrchestratorError /
// TransportError are converted to
// CoreError at the FFI boundary via `From` impls (ffi-surface.md
// §8.1): map-parse → Config,
// tun-attach → BadFd, auth-revoked → Auth, leak-unprotectable →
// HostFatal, else → Internal.

```

Classification: HostFatal (the core refuses to run a tunnel it cannot leak-protect) → the orchestrator emits Down{host-fatal} (§8) and start() returns Err. AlreadyStarted/NotStarted are caller-discipline errors surfaced as Dart exceptions, never silent no-ops (§11.4.6). A caught panic becomes Internal (a bug, surfaced honestly); with panic=abort (§3.1) an uncatchable panic aborts the process rather than corrupting the C-ABI.

10.2 Edge cases (each maps to a §12 test point)

#	Edge case	Required behaviour	Source
C1	start() called twice	second returns Err(AlreadyStarted); one live orchestrator, one tunnel (§6.1)	§4.3
C2	attach_tun(fd) on a closed/foreign fd	Err(BadFd); no half-open pump; kill-switch unchanged	§6.2
C3	status_stream receiver wedged > 64 events behind	Lagged → skip-to-latest; core unaffected (§5.3)	[v02-orch §4.2]
C4	iOS NE memory spike during upload soak	bounded buffer pool (§3.3); G3 soak must hold ≥30% headroom; else D-CLIENT-1 fallback	[research-ios_android §1]
C5	apply_map with malformed JSON	Err(Config); last-applied map stays live (fail-static, §7.4)	§7.2
C6	Revoke (peers_removed = active gateway)	tear session, close kill-switch, FFI Down{auth-failed}; < 1 s (§7.3/§8.5)	[v02-orch §6.4]
C7	Lagged/Closed on the FFI pump		§5

#	Edge case	Required behaviour	Source
		Lagged→continue; Closed→end Dart stream (§5.2)	
C8	Rust panic! in an FFI body	caught → CoreError::Internal; uncatchable → panic=abort process abort (never unwind across C-ABI)	§3.1/§4.3
C9	stop() then start()	clean re-arm: ladder reset to top, kill-switch re-armed closed (L-I8)	[v02-orch §5.2]
C10	frb-crate ≠ Dart-dep version	build-blocking drift finding (§12 UT)	[research-flutter_ffi §1]
C11	Android socket not protect()ed	tunnel loops its own packets back into tun — MUST protect every outbound transport socket	[research-ios_android §3]
C12	HarmonyOS/ Aurora VPN API absent in the fork	honest §11.4.3 SKIP- with-reason + tracked migration item; never a faked PASS	[research-flutter_ffi §4/§5]

11. Config knobs (CoreTunables)

The FFI passes through the orchestrator tunables [v02-orch §11]; the client-core adds the FFI/runtime knobs. Every numeric default tagged "derived" is the **spec recommendation**, calibrated against Phase-0 gates with captured evidence before it is frozen (§11.4.6 — a number is a guess until measured).

```
pub struct CoreTunables {
  pub orch: helix_core::Tunables,           // [v02-orch §11]:
    tick/handshake/keepalive/backoff/status
  // --- FFI / runtime ---
  pub tun_read_buf: usize,                  // bounded to
    negotiated MTU + headroom; default 1500+64 (§3.3)
    [research-ios_android §1]
  pub status_initial_emit:
    bool,                                   // emit Disconnected on subscribe; default
    true (C12, §5.2)
```

```

    pub ios_mem_sample_period: Duration, // phys_footprint
      sampler cadence in NE; default 1 s (derived) [research-
      ios_android §1]
  }

```

UNVERIFIED: keepalive_timeout = 25 s mirrors WG's REKEY-after-no-recv window but the exact boringtun-surfaced value is a Phase-0 measurement item [v02-orch §11]. tun_read_buf is the single strongest iOS-memory lever and is measured on device, never assumed (§3.3).

12. Test points (§11.4.169)

Per §11.4.169 every workable item declares its required test types from the closed vocabulary [06-phase0-spike-wbs §0.4]; every PASS cites rock-solid captured evidence (§11.4.5/.69/.107), never config-only. The FfiTunnelStatus event trace

- the on-device RSS report are the captured evidence for the core's claims.

Test id	§11.4.169 codes	What it proves	R ca ev
HC-1	UT	the §5.1 projection map is total + correct: every core TunnelStatus → expected FfiTunnelStatus; Down{stopped}→Disconnected	ta un \$5 m (u
HC-2	UT	FFI ⇔ Dart mirror parity: the generated Dart FfiTunnelStatus/Shields/ExitOption match the Rust #[frb(mirror)] types byte-for-byte; frb-crate==Dart-dep version	ge fix dr bu
HC-3	UT	SecretBytes Debug redacts + zeroizes on drop; no FFI return embeds an endpoint/secret (F-I5)	re as gr pr
HC-4	IT	start() builds the orchestrator, attach_tun(fd) pumps real WG datagrams tun↔transport on the netns rig; curl reaches LAN host	ne bo pl to cu
HC-5	E2E+FA	full status arc Disconnected→Connecting→Handshaking→Connected{plain-udp,direct,rtt} driven ONLY by the FFI stream; N=3 identical traces (deterministic §11.4.50)	FI to w sc vi (\$

Test id	§11.4.169 codes	What it proves	R ca ev
HC-6	E2E	reconcile restart-free: apply_map edit → peer reachable, pid unchanged, MapApplied.restart_free==true	fil de ne re ho un (C
HC-7	E2E+PERF	revoke < 1 s: peers_removed=active gateway → session torn, FFI Down{auth-failed}, kill-switch closed	tin st 1 re
HC-8	SEC	kill-switch closes on drop — no plaintext egress while Reconnecting/Down; off-tunnel :53 dropped when Connected	bl m Co to un sh le
HC-9	MEM	G3 make-or-break: iOS NE extension RSS under the device ceiling with ≥30% headroom, 30-min soak, plain-UDP AND MASQUE separately; upload-spike soak holds	re In ph R
HC-10	BENCH	the §3.1/§3.2 build profile produces the recorded staticlib size; build-std reduction measured (not assumed)	pe .a C af
HC-11	SC (stress+chaos)	start/stop/attach_tun/detach_tun churn N×; reconcile-source SIGKILL mid-Connected → loops keep forwarding (fail-static); no fd leak, bounded RSS	≥ ite bo le pa
HC-12	CONC+RACE	concurrent FFI calls (status_stream listener + set_shields + apply_map) on one core: no torn state, no deadlock; cargo test --features loom on the handle cell	co ha lo
HC-13	IT	connector posture: bin/helixd --mode=connector + the Flutter Connector flavor drive the SAME core; gateway→connector→LAN forward + reverse, ACL default-deny	ne al re un D
HC-14	SEC	Android protect()/protectFromVpn applied to every outbound transport socket — no tunnel-loopback	ne ca ou tr

Test id	§11.4.169 codes	What it proves	R ca ev
HC-15	CH+HQA	a challenges/helix_qa bank entry scores PASS only on the captured status arc + leak-free capture + G3 RSS report	pa en H au se ev ga (\$

Each meta-test ships a paired §1.1 mutation (e.g. flip the §5.1 projection to map `Down{stopped}→Down` → HC-1 FAILs; emit `Connected` before opening the kill-switch → HC-8 FAILs; remove the `tun_read_buf` bound → HC-9 must FAIL on the upload soak) so the gates provably cannot bluff [04_P0 §8, §11.4.5].

13. Open decisions & cross-doc contracts

13.1 Decisions surfaced (options + recommendation, never silently resolved — §11.4.6/§11.4.66)

#	Decision	Option A	Option B	Recommendation
D-CORE-1	FFI binding generator	frb v2 (Dart) + UniFFI (Swift/Kotlin) + cbindgen (HarmonyOS/Aurora)	cbindgen C-ABI everywhere	A — frb's StreamSink is the mature status-stream path; UniFFI gives typed Swift/Kotlin; cbindgen where UniFFI is thin [research-flutter_ffi §2/§7]
D-CLIENT-1	iOS NE memory fallback (if G3 fails)	shrink buffers / cap QUIC windows in iOS build	move MASQUE off-device or split-core app-extension IPC	Gate on the G3 measurement (§3.3); prefer A if it reaches ≥30% headroom [04_P0 §6.4]
D-CORE-2	iOS build-std adoption	nightly build-std +	stable toolchain, [profile.release] only	A for the iOS target if the size win is needed to pass G3; B

#	Decision	Option A	Option B	Recommendation
		optimize_for_size (~43% smaller)		elsewhere (stable) [research- ios_android §2]

13.2 Cross-document contracts this document fixes

Contract	Fixed value	Consumed by
helix-ffi exported surface (start/stop/status_stream/exits/ set_exit/set_shields/advertise/ apply_map/attach_tun/detach_tun)	§4.1	doc 03 (Dart wrapper), every native shim
FfiTunnelStatus + its projection from the core 5-variant TunnelStatus	§5	doc 03 §3.2 (Dart mirror), [v02-orch §4.1] (must emit the 5 core variants)
TUN-fd C-ABI seam (attach_tun(fd) / tun_out / set_tun_writer / detach_tun)	§6.3	every per-platform shim (NE/ VpnService/wg-nt/ tun)
apply_map reconcile entry (RouteMap/MapDelta shape, restart-free)	§7	doc 02 (WatchNetworkMap source), [v02-orch §6]
Build/memory profile (opt-level=z·LTO·panic=abort·strip + iOS build-std; ≤~12-15 MB steady-state, G3)	§3	the iOS NE build, [04_P0 §6]
--mode=connector packaging (one binary + headless daemon, advertise)	§9	doc 03 §6 (Connector flavor), [v02-orch §9]

Sources verified

- 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md [04_UI] — §0 (is/is-not), §1 (melos layout), §2 (flavors), §4 (Riverpod), §5 (FFI surface), §6 (per-platform shim matrix + HarmonyOS/Aurora), §10 (budgets).
- 04_VPN_CLD/HelixVPN-Architecture-Refined.md [04_ARCH §5] — §5.1 (two-cores strategy + frb/UniFFI), §5.3 (shim matrix), §5.4 (connector headless), §5.5 (reuse map), §5.6 (size/memory, iOS-NE = Rust-not-Go), §5.7 (Web/Aurora caveats), §11 (repo layout).

- 04_VPN_CLD/HelixVPN-Phase0-Spike.md [04_P0] — §6 (iOS NE memory experiment G3 + fallbacks), §9 (FFI boundary G5 + Rust/Dart skeletons), §10 (static-map shape), §11 (graduation).
- v09-research/research-flutter_ffi.md [research-flutter_ffi] — §1 (frb v2 2.12.0 + StreamSink), §2 (UniFFI not Dart-production-ready), §3 (staticlib/cdylib packaging), §4 (Aurora fork), §5 (HarmonyOS SIG fork), §6 (Riverpod 3.0 StreamProvider pause/autoDispose), §7 (synthesis).
- v09-research/research-ios_android.md [research-ios_android] — §1 (NE memory: 50 MiB documented iOS 15-18, ~15 MB observed kills, target ~12-15 MB, do-not-hardcode), §2 (Rust size optimization: opt-level=z/LTO/panic=abort/ strip + build-std), §3 (Android VpnService detachFd/protect/JNI), §4 (cross-platform synthesis).
- Sibling Volume-2 docs v02-data-plane/orchestrator-and-state.md [v02-orch] (Orchestrator, the 5-variant TunnelStatus, the reconciler diff/apply, the kill-switch/DNS coupling, Mode::Connector) and v02-data-plane/transport-trait.md [v02-tx] (the Transport trait, SecretBytes, invariants I1-I5). [SYN] cross-document synthesis — §5 (client architecture), §6 (repo layout), §8/§9 (submodules, constitution bindings).

Constitution: §11.4.44 (revision header), §11.4.6/§11.4.66 (decisions = options + recommendation, UNVERIFIED never fabricated), §11.4.5/§11.4.69/§11.4.107/§11.4.159 (captured/window-scoped evidence), §11.4.10 (SecretBytes no-leak), §11.4.28/.29/.74 (decoupled reusable workspace, snake_case, flat submodule), §11.4.169 (mandatory per-item test types). SPEC-ONLY — describes what to build, does not build it.

End of nano-detail specification — Volume 4 (Clients), helix-core-rust.md. Pairs with v02-data-plane/orchestrator-and-state.md (the orchestrator this FFI drives), v02-data-plane/transport-trait.md (the carrier seam), and doc 03 (the Dart helix_core_ffi wrapper + TunnelPlatform shim contract above this surface). The helix-ffi surface (§4.1), the FfiTunnelStatus projection (§5), the TUN-fd C-ABI seam (§6.3), and the --mode=connector packaging (§9) are the frozen Phase-0 surviving contracts [04_P0 §11]; their implementations may evolve, their shapes may not.